

Allison W. Li

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Computational biologist skilled in dynamical systems and spatiotemporal modeling, explainable machine learning, and scalable research software engineering for complex biological systems.

EDUCATION

University of Washington, Ph.D. Computational Biology

Sep 2023 – Jun 2028 (expected)

GPA 4.0/4.0 · Advisor: Dr. Neda Bagheri

University of Washington, B.S. Computer Science & B.S. Biochemistry (double major)

Sep 2019 – Jun 2023

GPA 3.73/4.0 · Phi Beta Kappa

RESEARCH EXPERIENCE

Graduate Research Assistant, MODYLS Lab, UW Biology (Advisor: Dr. Neda Bagheri)

Sep 2023 – present

Project: Forecasting CAR T therapy outcomes in solid tumors with multiscale spatiotemporal models

- Design and analyze systems of coupled nonlinear ODEs (mass-action and stoichiometric formulations) spanning three interacting scales (environment, cell population, subcellular) to characterize emergent dynamics from first-principles rate laws
- Architect and develop [ARCADE](#), a large-scale, open-source stochastic agent-based simulation framework in Java engineered for reproducibility, extensibility, and performance across millions of interacting agents; maintain the repository with version control, code review, and automated tests
- Run high-throughput parameter sweeps and sensitivity analyses over a high-dimensional design space on AWS Batch and Slurm HPC to quantify how parameters drive outcomes and identify optimal configurations
- Perform statistical analysis and visualization of high-dimensional, time-series simulation outputs to extract signals, test hypotheses, and translate results into actionable design guidance

Outcomes: Poster presentations at [SEED 2026](#) and [EBRC 2024](#); publication in progress

Skills: Java, Python, HPC, AWS, Git, ODE modeling, agent-based modeling, data visualization

Graduate Research Assistant, AIMS Lab, Paul G. Allen School of CSE (Advisors: Dr. Su-In Lee, Dr. Jessica Young) Dec 2023 – present

Project: Identifying signals of Alzheimer's disease progression by isolating salient variables in single-cell transcriptomics

- Develop ADVISE (Alzheimer's Disease Variational Isolation of Signal Embeddings), a variational autoencoder framework that embeds single-cell profiles into latent variables and separates disease-related signals from biological background in large, noisy datasets (~2M cells)
- Detect gene and pathway signatures from latent variables with explainable AI techniques (expected gradients, Hotspot, GSEApY pathway analysis) to nominate candidate treatment targets
- Collaborate with the Young Lab (UW Medicine) to design gene-editing experiments that validate novel AD therapeutic targets

Outcomes: Poster presentation at [MLCB 2024](#); publication in progress

Skills: Python, PyTorch, scvi-tools, SciPy, AnnData, explainable AI, single-cell sequencing analysis, data visualization

Research Assistant, Bedford Lab, Fred Hutch VIDD (Advisor: Dr. Trevor Bedford)

Apr 2022 – Sep 2023

Projects: Phylogenetic modeling of norovirus adaptive evolution; multiple linear regression (MLR) fitness modeling of epistasis

- Developed MLR models assigning fitness to SARS-CoV-2 variant lineages and assessed epistasis by analyzing fitness differences between lineages
- Modeled norovirus adaptive evolution with phylogenetic, population, and migration statistical analyses, revealing genotype-specific adaptive mutation rates
- Developed Python, Snakemake, and Augur scripts for data wrangling, used R for visualization, and constructed phylogenetic trees from bulk RNA sequences with Nextstrain

Outcomes: Publication; contributed to [Nextstrain open-source software](#); built the [Nextstrain live norovirus tracking dashboard](#); poster presentation at the [Fred Hutch SURP Poster Session](#)

Skills: Python, Snakemake, R, Augur, Nextstrain, RNA sequencing analysis, phylogenetic analysis, data visualization, MLR modeling

Programmer, BIOFAB, Institute for Protein Design (Advisor: Dr. Eric Klavins)

Sep 2021 – Apr 2023

- Developed software pipelines for [Aquarium](#), a cloud-based platform for lab inventory tracking and wet-lab protocol standardization
- Tested, revised, and designed features to improve Aquarium's efficiency
- Collaborated with lab technicians to standardize protocols and refine the user interface

Skills: Rust, Python, Docker

Research Assistant, Behavioral Ecophysics Lab, UW Biology (Advisor: Dr. Alejandro Rico-Guevara)

Jan 2020 – Apr 2023

Project: Hummingbird thermoregulation through CT-scan segmentation

- Landmarked 1,000+ CT scans in SlicerMorph to build an extensive morphological database
- Ran statistical analyses of 3D hummingbird flight trajectories in MATLAB and Python

- Preprocessed CT scans for segmentation and statistical shape analysis in 3D reconstruction; wrote SlicerMorph protocols for future students

Outcomes: Poster presentation at the [2022 Undergraduate Research Symposium](#)

Skills: SlicerMorph, Python, MATLAB, scan segmentation and processing

ADDITIONAL EXPERIENCE

Global Programs Intern, SightLife

Nov 2020 – Sep 2021

- Built a patient-records database with partner LVPEI hospitals in India (R, Excel) and identified statistical trends pointing to barriers in access to corneal care
- Designed an ophthalmic technician curriculum (activities, wet labs, lesson plans) for international programs in collaboration with the US FDA and LVPEI

Skills: R, Shiny, Excel, Python, data visualization

PUBLICATIONS (* equal contribution)

Published / Accepted

- K. R. Andrew*, J. Chang*, C. Roemer, J. Hadfield, V. Lin, A. F. Brito, R. Daodu, I. A. Joia, K. Kistler, **A. W. Li**, L. H. Moncla, M. I. Paredes, D. Kühnert, L. M. Torres, L. Urbanska, I. Aksamentov, E. B. Hodcroft, J. Huddleston, J. T. McCrone, T. R. Sibley, J. S. J. Anderson, J. Lee, R. A. Neher, T. Bedford. Nextstrain automates real-time phylodynamic analysis of open data for endemic and emerging pathogens. *Bioinformatics* (accepted, 2026) [[Bioinformatics Article](#)]

In Preparation

- **A. W. Li***, W. Qiu*, C. M. Arian*, J. E. Young, S.-I. Lee. Disentangling context-specific Alzheimer's disease progression signatures through an explainable AI framework. *In preparation*
- **A. W. Li**, J. Y. Cain, N. Bagheri. In silico modeling forecasts parameter optimizations to improve resiliency of SynNotch-AND gate CAR T circuits against immunosuppressive environments. *In preparation*
- **A. W. Li**, J. Y. Cain, N. Bagheri. Spatially structured contexts limit iCAR selectivity despite improved migration, treatment parameters, and circuit design. *In preparation*

ORAL PRESENTATIONS

- In silico modeling guides design of combinatorial antigen recognition in engineered CAR T therapies [[PDF](#)]
A. W. Li, J. Y. Cain, N. Bagheri · *Synthetic Biology Consortium Automation Symposium 2026, Fred Hutch, Seattle, WA*
- Multiscale modeling informs synthetic CAR T circuit design for cell therapy [[PDF](#)]
A. W. Li, J. Y. Cain, N. Bagheri · *UW Biology Graduate Seminar 2026, Seattle, WA*
- Speaker [[video](#)]
Women in Machine Learning & Data Science (WiMLDS), Mar 2026

POSTERS & EXTENDED ABSTRACTS

- In silico modeling forecasts iCAR inefficacy in tumors and evaluates migration constraints on selectivity [[PDF](#)]
A. W. Li, J. Y. Cain, N. Bagheri · *Synthetic Biology: Engineering, Evolution & Design (SEED) 2026, Denver, CO*
- Modeling multi-scale in silico dynamics of combinatorial recognition circuits in T cell therapies [[PDF](#)]
J. Y. Cain, **A. W. Li**, N. Bagheri · *Engineering Biology Research Consortium (EBRC) 2024, Seattle, WA*
- Capturing biological signals in Alzheimer's single-cell transcriptomic data by isolating salient variations [[PDF](#)]
A. W. Li, W. Qiu, E. Weinberger, S.-I. Lee · *Machine Learning in Computational Biology (MLCB) 2024, Seattle, WA*
- Phylogenetic modeling analysis of norovirus reveals varying genotype and gene adaptive mutation rates [[PDF](#)]
A. W. Li, J. Huddleston, K. Kistler, T. Bedford · *Fred Hutch SURP Conference 2022, Seattle, WA*
- Poster presentation, Behavioral Ecophysics Lab research
A. W. Li · *UW Undergraduate Research Symposium 2022, Seattle, WA*

FELLOWSHIPS, AWARDS & FUNDING

Washington NASA Space Grant Fellowship, Washington NASA Space Grant Consortium

Jun 2026

Top Scholar Fellowship, UW Department of Biology

Sep 2023

Tapia Conference Scholarship, Paul G. Allen School of CSE

Jun 2022

Burkhardt Family Scholarship, Paul G. Allen School of CSE

Sep 2021

Phi Beta Kappa Honor Society, Member

Sep 2020

TEACHING & MENTORING

- Teaching Assistant**, UW Department of Biology Dec 2023 – Mar 2024
- Led weekly wet-lab sections (pipetting, gel electrophoresis, PCR, dialysis), graded assignments and exams, revised exam questions, and held office hours for Introduction to Biochemistry (BIOL 200)
- Teaching Assistant**, UW Computer Science & Engineering Sep 2021 – Jun 2022
- Led weekly sections, graded assignments, wrote exam questions, and held office hours for Software Design and Implementation (CSE 331) and introductory Java programming (CSE 142)
- Peer Educator**, UW Interdisciplinary Honors Program Dec 2019 – Mar 2020
- Wrote lesson plans and curriculum supporting incoming first-year students academically and in health and wellness; mentored 13 students

SERVICE & OUTREACH

- Scholarship Review Panel**, UW Biology Feb 2026 – present
- Score and evaluate funding proposals using NIH grant review criteria
- Art Director**, Grey Matters Neuroscience Journal Mar 2021 – Apr 2024
- Managed artists through each publication cycle: set timelines, ran illustration workshops and critiques, and hired new artists
 - Founded UW's first [neuroscience art gallery](#) to promote interdisciplinary thinking and diversity in research; partnered with student-run art organizations
 - Illustrated [journal articles](#) ([portfolio](#))
- Volunteer Chair**, UW Manana Sep 2020 – Jun 2023
- Partnered with Solid Ground to host in-person and virtual celebrations for at-risk youth; organized fundraisers; recruited, screened, and trained 10+ volunteers

MEDIA

- [Women in Machine Learning and Data Science Presenter](#) Mar 2026
- [Nextstrain blog: announcement of norovirus release](#) Jan 2026

SKILLS

Languages: Python, Java, R, MATLAB, Rust · **ML & data:** PyTorch, scvi-tools, AnnData, SciPy, GSEAPy, explainable AI
Modeling: ODEs, agent-based models, sensitivity analysis, phylogenetics · **Infrastructure:** Git, Docker, Snakemake, AWS, Slurm
Imaging & lab: SlicerMorph, CT segmentation, PCR, gel electrophoresis